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Streamflow response to catastrophic windthrow and forest recovery in subalpine spruce forest

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ABSTRACT

Windstorms account for about half of forest disturbances in Europe and are considered as drivers of natural regeneration of plant species, accompanied by significant changes in hydrological and soil moisture conditions. Our study aims to identify the effects of deforestation caused by catastrophic windthrow on streamflow characteristics in conjunction with meteorological and soil moisture data. The research was conducted in the headwater catchment (34.4 km^2), in the Western Tatra Mountains, Poland. Streamflow characteristics such as average, high, and low flows, coefficient of variability, as well as flashiness of the streamflow (Richard-Baker Flashiness Index) and drought frequency (Standardised Streamflow Index) were calculated in the pre-event (2007–2013) and post-event (2013–2020) periods. The analysis was carried out for calendar months, seasons of the year, as well as for the growing and non-growing seasons. Wavelet Analysis was applied to track the changes in streamflow time series in relation to meteorological conditions (precipitation, air temperature) and remotely sensed soil wetness in the root zone. Principal Components Analysis was applied in order to identify the major drivers of the observed variability. The results revealed increasing low (+14%), average (+8%), and maximum monthly flows (16%) in the post-event period. The most significant increase in low flows (+47%, $p<0.1$) occurred during snowmelt. The increase in average (+31%, $p<0.1$) and high flows (+90%, $p<0.05$) occurred during mid-winter thaws. The frequency of hydrological droughts decreased by up to 36% in the post-event period, despite similar climatic conditions. The flashiness of streamflow increased during mid-winter thaws and rainfall events. The short-scale periodicity in streamflow increased in the post-event period. The results revealed the presence of 3-month periodicity in the root zone soil wetness associated with long-term soil water storage, which decreased after the catastrophic windthrow. Streamflow in the post-event period became strongly coupled with precipitation and root zone soil wetness at short period bands (~3–16 days) in hydrologically active season. Short-scale correlation between precipitation and streamflow was decreasing with the gradual succession of vegetation.

1. Introduction

Forests occupy one third of European territory and provide more than 4 km^3 of water each year to European citizens, providing more than 4% of European renewable water resources (EEA, 2015). In Poland, forest areas cover approximately 30% of the territory, making it as a medium-forested country. Massive deforestation took place in the XIX and XX centuries and was related to industrialisation and urbanisation. After the Second World War, forested areas accounted barely 20%, therefore, further forest management strategies relied largely on afforestation with tree monocultures. These monocultures hardly adapted to the places where they were planted and were more prone to diseases that caused further degradation of the tree stands. Therefore, for almost a century, natural disturbances, such as storm events, were one of the main causes of deforestation in Central Europe (Fischer et al., 2002), including Poland. Previous research in the region reported the consequences of deforestation of different origins such as strong foehn winds in mountain areas (Sajdak et al., 2021; Siwek, 2021; Żelazny et al., 2019a), massive bark beetle infestation (Hesslerová et al., 2018), acid rains (Bałazy et al., 2019) and forests wildfires (Kundzewicz and Matczak, 2012). In some cases, these changes were irreversible, inhibiting the natural regeneration of the windbreak fields.

Tracts of windbreaks and defoliated trees constitute an inseparable part of Polish mountain protected areas, occurring both in the

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Carpathian and Sudeten Mountains. Unfortunately, the area of damaged trees has grown rapidly in the past decade, which is associated with climate changes and a higher frequency of catastrophic events (Feng et al., 2023; Pociask-Karteczka et al., 2018). Such changes may, in turn, bring far-reaching implications to the water balance at catchment or even basin scale.

Forest canopy plays a critical role in the hydrological cycle by regulating a wide range of hydrological processes and fluxes at surface and subsurface level, thus modulating terrain water storage and the volume of water that feeds groundwater, rivers, and lakes (Knighton et al., 2020). At the same time, forest ecosystems consume the most water among other plant communities (Bastrup-Birk and Gundersen, 2004; Calder et al., 2003). Therefore, forest disturbances caused by catastrophic events or human activity are likely to affect the rate and magnitude of hydrological processes. Previous research showed that deforestation causes several changes in the hydrological cycle, often bringing contradictory, nonsignificant, or unexpected results. The impact of deforestation on the water cycle is complex and depends not only on the area affected by deforestation, but also on climatic conditions, soil types, and tree species. Failure to detect substantial change in runoff could also result from the scale disproportion between the treatment area and the recharge area of the stream (Biederman et al., 2015). The effects of deforestation are more evident in areas dominated by coniferous forests, since they retain approximately 10% more water than broadleaved forests or mixed forests (EEA, 2015); as well as in semi-arid zones due to the large deficit between precipitation and potential evapotranspiration (PET) (Huxman et al., 2005). The most recent synthesis on the hydrological sensitivity of catchments to forest changes (Hou et al., 2023) reported, however, that the hydrological sensitivity to forestation is significantly larger than to deforestation and is more evident in Lithosol-dominated catchments with low water storage and pure forest specie.

Most studies on the impact of deforestation on hydrological conditions address its effects on flow characteristics (i.e., baseflow, low, average, and peak flows), as well as changes in water storage in the snow cover and soils resulting from infiltration, evapotranspiration, and interception. Previous studies reported that deforestation causes an increase in streamflow and annual runoff (Costa et al., 2003; Sørensen et al., 2009). About 55% deforestation significantly alters the hydrological characteristics of the catchment and increases discharge by about 20% (Coe et al., 2011). Research suggests that the removal of trees from riparian areas has an even greater impact on forest retention capacity than deforestation done away from the river network (Abdelnour et al., 2011; Boggs et al., 2016).

Changes in rainfall-induced peak flows in deforested catchments show different responses. Both increases and decreases in maximum flows were observed. Several studies reported the increase of storm runoff (Goodbrand et al., 2022) and the magnitude and frequency of peak flows (Hlásny et al., 2015; Kuras et al., 2012; Liu et al., 2015). Studies suggest that the impact of forest disturbances on peak flows can differ between rainfall, rain-on-snow, and spring snowmelt events (Winkler et al., 2010). Increasing peak flows are frequently reported in snow-dominated catchments (Grant et al., 2008). Research suggests that a highly variable peak flow response to deforestation could be related to the role of transient snow cover and the percentage of canopy removal (Hudson, 2001). Other studies, however, suggest that the impact of forest disturbances on hydrology is more evident in rainfall-dominated catchments due to a strong relation between peak flows and storm events (Moore and Wondzell, 2005).

Changes in forest cover have a significant impact on average and low discharges in the dry season, as well as on baseflow and groundwater recharge (Cheng et al., 2017). Research suggests that clear-cut above 60% causes an increase in average and low flows by up to 50% and results in a significant increase in low and average flows in the dry season (Zhang and Wei, 2012). Increasing runoff from forest disturbed catchments is often attributed to decreasing evapotranspiration and interception (Coe et al., 2009; Costa et al., 2003). The research showed that forest logging could decrease evapotranspiration by up to 25–40 % (Wei et al., 2022), while the interception of snow in boreal forests can reach up to 60 % (Pomeroy and Schmidt, 1993). The intensity of interception and evapotranspiration can, in turn, create a soil moisture deficit during the growing season, affecting surface waters and energy fluxes in the soil root zone, so-called rhizosphere (Stocker et al., 2023). This zone extends up to 100 cm and stores the water accessible to plant roots (Baldwin et al., 2017; Jackson et al., 2005). Research suggests that soil moisture is an important factor controlling rainfall-runoff processes, which in turn results in different streamflow patterns (Evaristo and McDonnell, 2019; Penna et al., 2011).

Previous research on deforestation in the Polish Tatra Mountains reported mostly changes in stream water chemistry (Sajdak et al., 2021), bedload transport (Płaczkowska et al., 2020), sediment transport (Strzyżowski et al., 2021), and soil temperature (Siwek, 2021). So far, no attention has been paid to the accompanying hydrological changes. Despite the strong psychological effect of broken trees, research suggests that natural disturbances can serve as factors of natural regeneration of forests that allow restoration of native tree species. This, in turn, may be favoured by greater availability/limitation of water. Therefore, further studies on forest hydrological disturbances are necessary to develop strategies for windfall and natural resources management in mountain protected areas.

Given the above, the objective of this study is to identify changes in streamflow characteristics and periodicity related to deforestation in the headwater mountain catchment in the temperate climate zone that has been artificially dominated by spruce monocultures. In principle, we investigated the impact of deforestation on 1) streamflow characteristics, 2) frequency of hydrological droughts, 3) flashiness of the stream, 4) periodicity in streamflow and its relation to meteorological conditions and soil wetness in the root zone. In contrast to other studies carried out in this field, analysing only streamflow changes at regional scale using a paired-catchment approach, our study focuses on a single headwater catchment and analyses streamflow changes, including hydrological drought frequency and streamflow flashiness, in relation to rainfall and soil–water storage in the root zone using a paired-year approach. We believe that this approach will provide a more detailed explanation of the hydrological changes driven by the catastrophic windfall.

2. Study site

The studied catchment of the Kościeliski stream is located in the Western Tatra Mountains (Fig. 1). The Tatra Mountains represent the highest mountain range in the Carpathian Mountains situated on the Polish-Slovak border. The Tatra Mountains are protected within national parks, which are jointly designated as a transboundary UNESCO biosphere reserve.

The Tatras are characterised by a complex geological structure. The Kościeliski catchment lies within all the main tectonic units of the Tatra Mountains. The highest parts of the catchment are built of crystalline and metamorphic rocks, while the lowest parts are built of carbonate rocks such as limestone, dolomite, and marls that form several nappes (Piotrowska et al., 2015). High-elevated southern part of the catchment is characterised by glacial and periglacial relief, while the northern part of the area has typical fluvio-denudational and karst relief. Soils are dominated by Cambic-Rendzic leptosols and Eutric Cambisols with depths of 0.5 m and 1 m, respectively (Skiba, 2002). The matrix is composed of silty loam with a changeable content of cobbles and gravels that increases with depth.

In terms of hydrological conditions, the Tatra catchments are characterised by the largest runoff in the country of about 50 $dm^3 s^{-1} km^{-2}$. The hydrological regime of the Tatra streams is characterised by one flood season resulting from spring snowmelt prolonged by summer rain (Żelazny et al., 2019b). Low flows prevail mostly in winter, when much

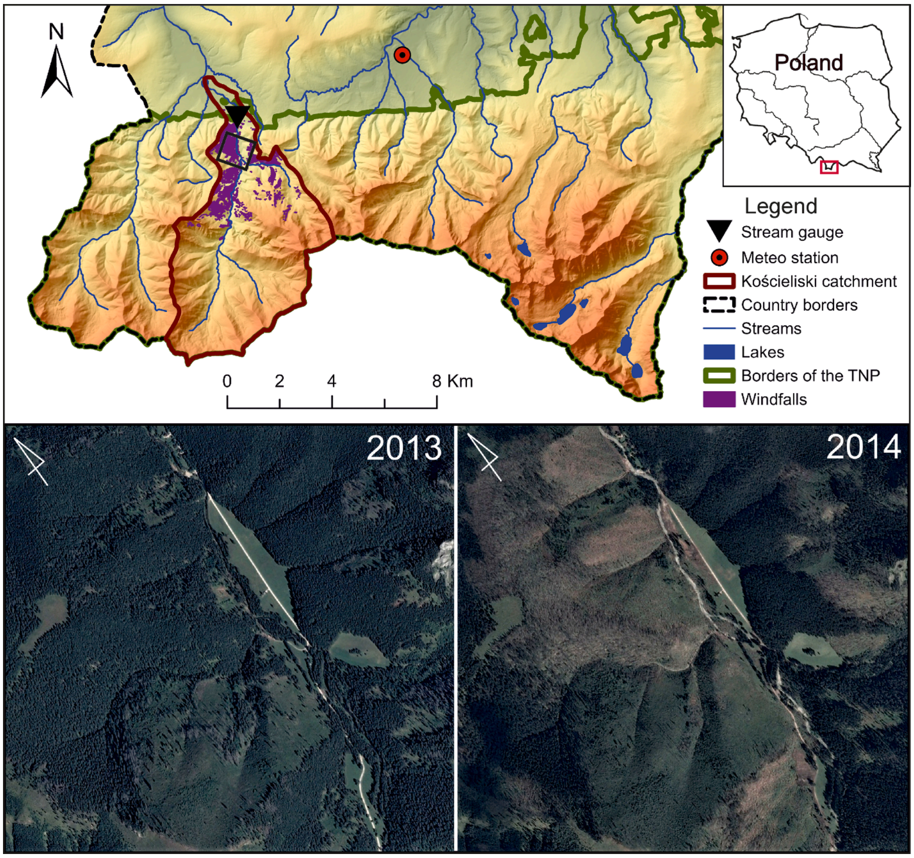

Fig. 1. Study site location. Aerial images from 2013 and 2014 presenting a downstream reach of the Kościeliski stream with deforested slopes (© Google Earth, Maxar Technologies 2023).

of the precipitation is stored in snow cover, and streams are recharged by karst springs. The Kościeliski stream is a perennial stream of about 9 km in length. The area of the catchment is approximately 34.5 km^2 with an average slope of about 30°.

According to the Köppen climate classification, the Tatra Mts. belong to temperate climate zone and are characterised by the altitudinal zonation of climatic conditions and vegetation zones. The average annual air temperature varies between + 6.0°C and 2.0°C, while the annual sum of precipitation ranges between 1120 mm at the foothills up to 1800 mm at the mountain peaks (Niedźwiedź, 1992). The elevation in the Kościeliski catchment extends between 920 and 2158.5 m a.s.l. thus several climatic belts can be distinguished within the catchment starting from the moderately cool (1150–650 m), followed by cool (1550–1150 m), very cool (1850–1550 m), and moderately cold (2200–1850 m). The highest parts of the Tatra Mts. (>2200 m) belong to the cold climatic belt (Niedźwiedź, 1992).

The vegetation cover in the lowest parts is represented by mixed montane forest, followed by subalpine spruce forest, dwarf pine belt, and alpine meadows ending with the uppermost sub-nival zone dominated by bare rocks. The forest of the Kościeliski catchment is, however, artificially dominated by Norway spruce monocultures, which in fact, are more susceptible to winds owning to their shallow root system. Additionally, spruce monocultures are also vulnerable to bark beetle infestation, posing another threat to forests in this area (Grodzki and Fronek, 2017).

Tree stands of the Tatra National Park have been repeatedly damaged in recent 100 years by windstorms, some of them being catastrophic. Between 23 and 26 December the Polish Tatra Mountains, were seriously hit by hurricane-force winds, with a maximum wind speed on 25 December reaching up to 200 km h^{-1}. Winds destroyed more than 100,000 m^3 of trees while the wind damaged area amounted approximately 2.25 km^2 (~6% of the catchment area). The damaged area is large given that forested areas (below the tree line ~1500 m a.s. l.) occupied approximately 52% of the catchment. Tree damage occurred in the active and passive protection zone. According to the legislation, broken trees in the active protection zone were removed, while trees in the passive protection zone were left. Approximately 22,000 m^3 of broken and fallen trees was removed in 2014. In the following year, the spruce bark beetle began to colonise the survived spruce trees in the vicinity of the windbreak (Grodzki and Fronek, 2017).

3. Methods

3.1. Data

The data include time series of daily streamflow of the Kościeliski Stream encompassing the period of 14 years from December 2007 to December 2020. The time series were divided into the pre-event (Dec 2007 – Nov 2013) and post-event (Dec 2013 – Nov 2020) periods, having the same length (7 years). The meteorological time series, including daily air temperature (min, max, average) and precipitation totals, were obtained from the synoptical station in Zakopane, located approximately 6 km from the water gauge. Hydroclimatic data (Q, P, T) comes from the national monitoring network of the Institute of Meteorology and Water Management (IMGW). Daily values of root zone soil wetness up to 100 cm, representing the maximum thickness of soil in this area, at deforested slopes (Fig. 1) in the Kościeliska Valley were obtained from the NASA POWER reanalysis data (MERRA-2), with a resolution of 0.713 km (https://power.larc.nasa.gov/). Variability in streamflow was analysed for months and seasons of the calendar year (winter: Dec-Feb, spring: Mar-May, summer: Jun-Aug, autumn: Sep-Nov), as well as for the growing (Apr-Oct) and non-growing seasons (Nov-Mar). The growing season refers to months with an average air temperature above 5°C.

3.2. Analysis

Characteristics of streamflow and meteorological time series, such as average, median, minimum, and maximum values, as well as the coefficient of variability (Cv), were calculated for corresponding aggregation periods. The significance of average and variance between the pre- and post-event characteristics of streamflow was tested at significance levels of 0.05 and 0.1 using ANOVA and Scheffe test correspondingly.

Streamflow variability in the pre- and post-event periods was also characterised in terms of the Richard-Baker Flashiness Index (RBI) and Standardised Streamflow Index (SSI). The Standardised Streamflow Index (SSI) is commonly used to assess and characterise hydrological droughts. The index was developed by Vicente-Serrano et al. (2012) and is analogous to the Standardised Precipitation Index (SPI) (McKee et al., 1993), which has been typically used to characterise meteorological droughts. The resulting SSI and SPI consist of negative and positive values, representing drought/dry (SSI, SPI<0) and non-drought/wet (SSI, SPI>0) events, respectively. The calculations of both indices require the selection of the most suitable probability distribution function to the variability of streamflow (SSI) and precipitation (SPI) over the analysed time domain. In contrast to SPI, which is widely based on gamma distribution (McKee et al., 1993), no single probability distribution has been recommended for SSI (Vicente-Serrano et al., 2012). In this study SSI index was calculated using three distributions (Gamma, Pearson III, and log-Logistic), of which gamma provided the most adequate fit. Therefore, for further analysis, the SSI and SPI indices were calculated over 1-month aggregation periods (SSI-1, SPI-1) using gamma probability distribution. The resulting values were categorised using the classification criteria of SPI (Table 1), recommended by the World Meteorological Organisation (Svoboda et al., 2012).

Changes in streamflow flashiness related to deforestation were evaluated based on the Richard-Baker Flashiness Index (RBI), which is defined as the mean of the absolute changes in runoff divided by the average runoff (Baker et al., 2004). The index is commonly used to assess the frequency and rapidity of short-term changes in streamflow related to land use changes that can lead either to an increase or decrease in flashiness. RBI values in the pre- and post-event periods were calculated for 1-month aggregation periods.

The Principal Components Analysis (PCA) was applied to identify independent variables that shape the variability of streamflow in relation to abiotic components of the physical environment. Component loadings in the PCA analysis were calculated based on 11 hydroclimatic variables including: Q_{avg}, Q_{min}, Q_{max}, T_{min}, T_{max}, T_{avg}, P, RBI, SSI, Coefficient of Variability (Cv), and Quartile Variation Coefficient (V). The number of the most important factors was determined based on the Kaiser criterion. Factor loadings ≥0.5 were used to indicate significant correlations.

The periodicity in the streamflow time series was analysed using the Continuous Wavelet Transform (CWT). In the transformation, the Morlet wavelet was applied owning to reasonably good localisation in the time and frequency domain (Torrence and Compo, 1998). The covariance between streamflow and meteorological time series (precipitation, air temperature) and soil wetness in the root zone was analysed using the Wavelet Coherence function (WTC). The WTC describes a local correlation between two CWTs for a given time scale. The WTC shows not only the period of common oscillations, but also the phase of oscillations indicated by the arrow direction. Right-pointing arrows indicate that two signals are in-phase (i.e., the phase shift between two time series is $\alpha=0$), bottom-pointing arrows indicate that first series is leading the second by 90°, (i.e., the phase shift is $\alpha=\pi/2$), left-pointing arrows indicate that two series are in antiphase (i.e., the phase shift is $\alpha=\pi$). The up-pointing arrows indicate that Y is leading X by 90°. The significance of periods in CWT and WTC was tested against the red noise power spectrum at the 5% significance level using the Monte Carlo method. The Wavelet Analysis was performed in MATLAB software based on the script developed by Grinsted et al. (2004).

Table 1
Drought classification based on the Standardised Streamflow Index values.

SSI value	Description
SSI ≥ 0.0	Non-Drought
−1.0 ≤ SSI < 0.0	Mild Drought
−1.5 ≤ SSI < -1.0	Moderate Drought
−2.0 ≤ SSI < -1.5	Severe Drought
SSI ≤ -2.0	Extreme Drought

Note. SSI –Standardized Streamflow Index.

4. Results

4.1. Changes in streamflow characteristics

Monthly low flows increased significantly by about 140 dm^3s^{-1} (+14%, p=0.07) in the post-event period (Fig. 2, Table 2). The highest increase in low flows (+17%, p=0.04) was observed in the growing season than in the non-growing season. On a seasonal time scale, the highest increase in low flows occurred in autumn (+12%, p=0.06). However, the most significant increase in low flows was observed in April (+27%, p=0.08), which overlaps with the snowmelt period. A significant increase in high (+90%, p=0.02) and average flows (+31%, p=0.07) occurred in February (Fig. 2) during mid-winter thaws and the highest thickness of snow cover. In the remaining months, changes in the pre- and post-event periods were not significant (p>0.1). A noticeable increase in the magnitude of peak flows during snowmelt (April-May) and rainfall events (July) was observed in the post-event period (Fig. 2), however, this increase was not statistically significant. No significant changes in average and high flows (p>0.1) were observed in seasons of the calendar year and in the vegetation seasons (Fig. 2).

4.2. Changes in drought characteristics

In the pre-event period, hydrological droughts, as indicated by SSI, occurred more frequently in the growing season than in the non-growing season (Fig. 3A). The number of meteorological droughts in the pre-event and post-event period was comparable (Fig. 3C). The frequency of meteorological droughts (SPI<0) decreased by less than 5% in the post-event period, while the frequency of hydrological droughts (SSI<0) decreased by up to 36% in the post-event period (Fig. 3C). The highest decrease in the frequency of drought between the pre- and post-event periods was observed mainly for mild-drought events (Fig. 3A). The monthly SSI values increased significantly in the post-event period (p=0.04), indicating less severe and less frequent drought events. For example, in the pre-event period, the median values of SSI in winter, spring, and autumn ranged between -0.5–0 indicating the dominance of mild-drought events, while in the post-event period, the median values of SSI in these seasons increased above 0 indicating the dominance of non-drought events (Fig. 4B). The highest increase in SSI values was observed in the non-growing season (p=0.07) than in the growing season (Fig. 4C). On a monthly basis, the frequency of hydrological droughts decreased in the post-event period in all months except June (Fig. 3B). However, only for February was the increase in SSI significant (p=0.07) (Fig. 4A). No significant changes were observed between the seasons of the calendar year (Fig. 4B).

4.3. Changes streamflow flashiness

The mean RBI values for daily flows in the post-event period were approximately 13% higher (RBI=0.17) than in the pre-event period

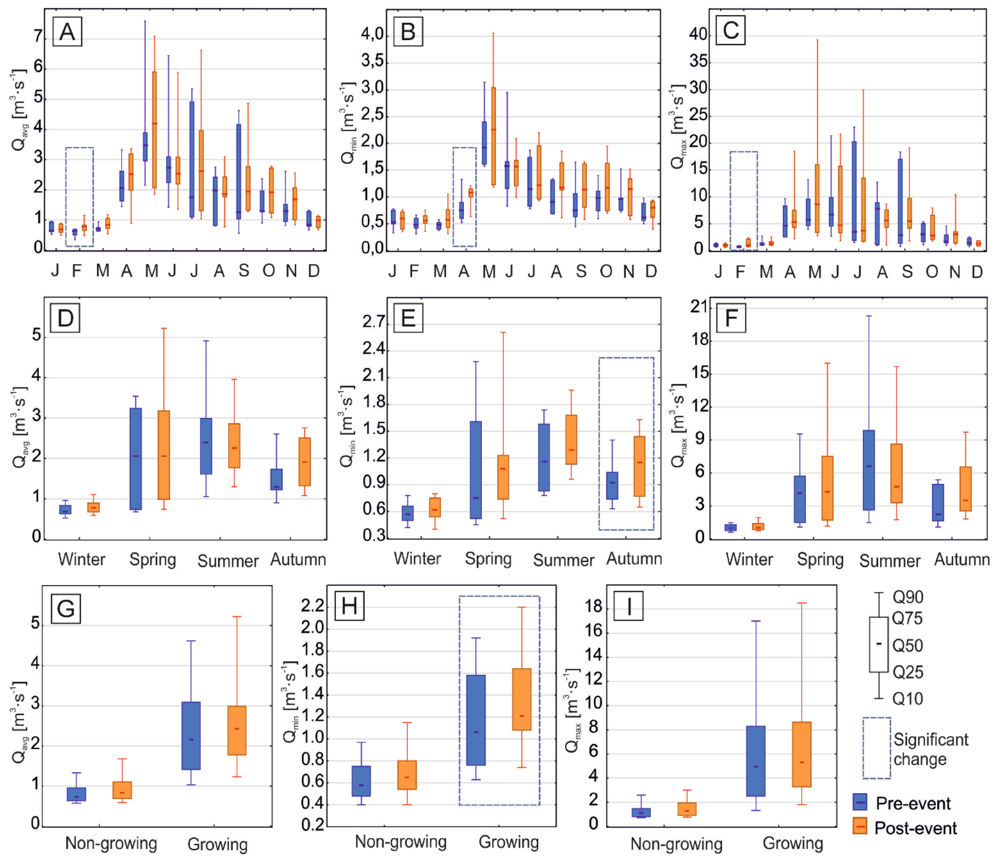

Fig. 2. Boxplots presenting positional measures of the main streamflow characteristics at monthly and seasonal time scales in the pre- and post-event periods.

Table 2
Summary of streamflow statistics in the pre- and post-event period.

	Q_{avg} [m^3s^{-1}]	Q_{max} [m^3s^{-1}]	Q_{min} [m^3s^{-1}]	SSI-1 [-]	RBI [-]
Period	Pre-event				
Avg	1.81	4.45	0.98	−0.13	0.12
Me	1.29	2.43	0.78	−0.27	0.10
Min	0.35	0.52	0.31	−2.00	0.01
Max	7.59	23.00	3.14	2.12	0.39
Period	Post-event				
Avg	1.96	5.18	1.12	0.16	0.13
Me	1.60	2.80	1.05	0.25	0.10
Min	0.48	0.58	0.31	−2.42	0.02
Max	7.09	39.30	4.06	2.29	0.43

Note. Avg – average, Me – Median, Min – minimum, Max – maximum.

(RBI=0.15), suggesting higher variability of daily streamflow (i.e., steeper rising and falling limbs). The monthly RBI values in the pre- and post-event period have not differed significantly (p>0.1). The monthly RBI values in the growing season increased in the post-event period, however, the increase in RBI in the growing season was not significant (+17%, p>0.1). In the non-growing season, RBI values were comparable (Fig. 5C). Changes in RBI values in seasons, as well as in the growing and non-growing seasons were not significant (p>0.1) (Fig. 5B-C). However, on a monthly basis, a significant increase in RBI values was observed in the post-event period in September (+58%, p=0.098) and in February (+73%, p=0.079) (Fig. 5A). The increase in streamflow flashiness in September coincides with late summer frontal rainfalls and low soil wetness after prolonged summer droughts. On the other hand, the increase in streamflow flashiness in February coincides with mid-winter thaws, which plausibly are favoured by faster snowmelt in sun-exposed windbreak fields.

4.4. Factors controlling streamflow variability

The PCA analysis showed that streamflow variability in the pre-event period was shaped by two independent factors (Table 3). The first factor (F1) explained the maximum variance (62%) in the data set and was related to the hydrological regime of the stream, i.e., the periodicity of recharge sources such as snowmelt in spring, rainfalls in summer, and groundwater recharge in winter. The second factor explained 16.4% of the observed variability and was related to climatic conditions.

In turn, in the post-event period, three factors were responsible for the observed variability in the data set (Table 3). The first two factors (hydrological regime and meteorological conditions) were the same as in the pre-event period, explaining a similar percent of variability (61.9% and 17.3%, respectively). The third factor (F3) explained approximately 10.3% of the observed variability and was apparently associated with deforestation. It described the relation between low flows and SSI index, i.e., the lower the monthly low flow (F_L: Q_{min}=-0.58) the lower the SSI index (F_L: SSI-1=-0.52). Although the relationship between low flows and the frequency and severity of hydrological droughts is obvious, it may be surprising that this relationship has been clearly distinguished only after deforestation. This might serve as evidence of streamflow changes related to deforestation.

4.5. Changes in streamflow periodicity

The streamflow time series are characterised by strong annual periodicity, as indicated by a red horizontal band concentrated at ~365 days in the wavelet power spectrum (Fig. 6). The largest flood events that occurred between May and June 2010 and in May 2014 are visible as elongated red lobes. A considerable lack of short-scale periodicity is observed in winter due to reduced streamflow caused by freeze-up and the increased retention of water in snow cover. The accompanying low

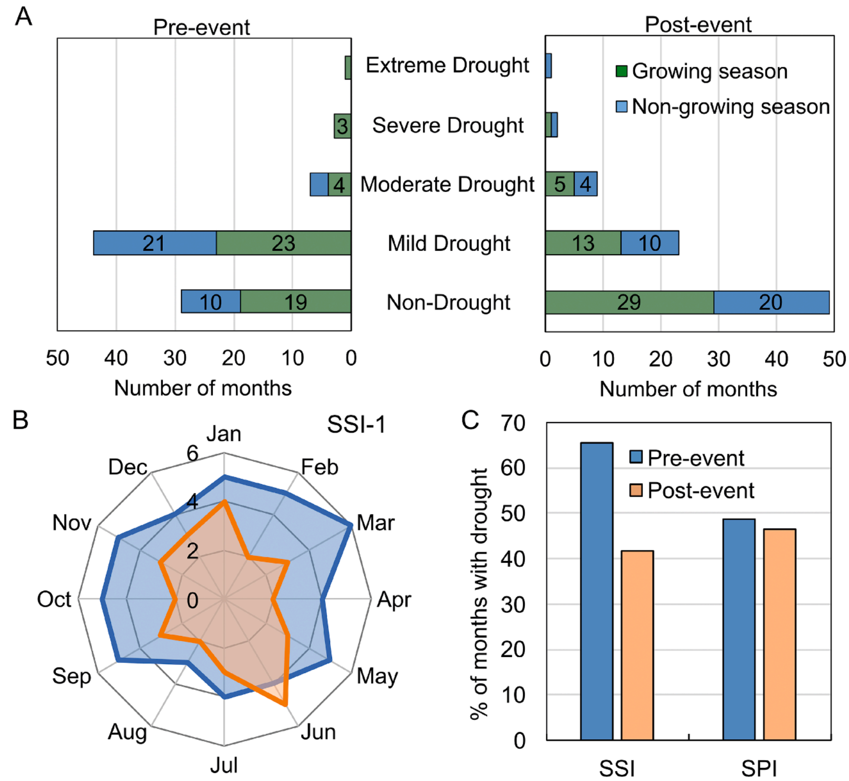

Fig. 3. Number of droughts and non-drought months in the pre-event and post-event periods (A), frequency of droughts in months (B), % of months with hydrological (SSI) and meteorological (SPI) droughts in the pre-event and post-event period (C).

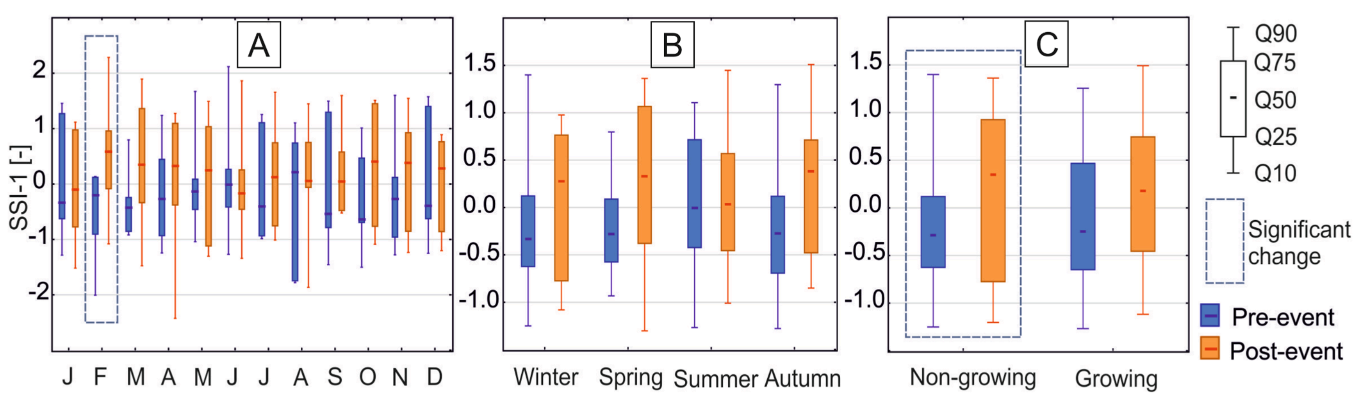

Fig. 4. Variability of the Standarised Streamflow Index (SSI-1) in the pre- and post-event period.

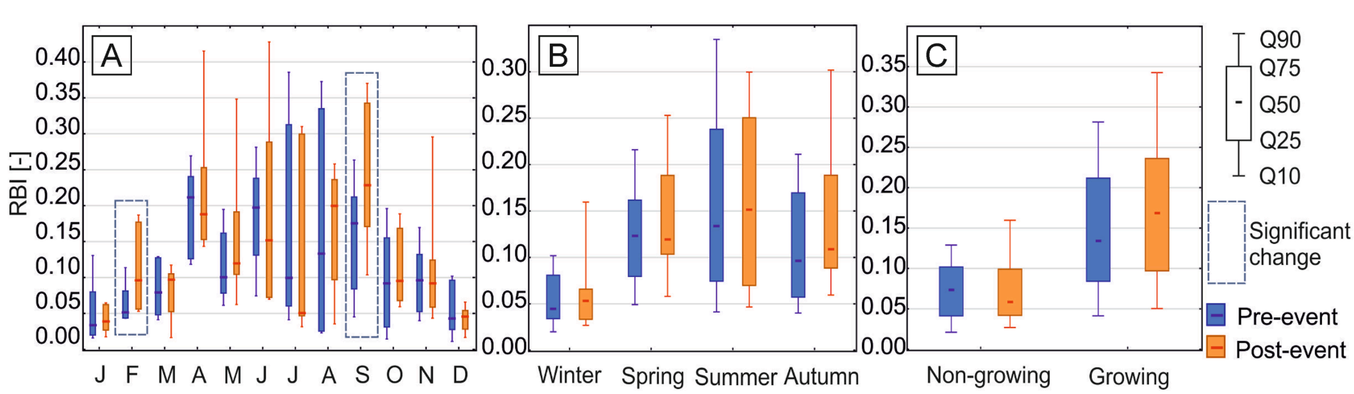

Fig. 5. The variability of the Richard-Baker flashiness Index (RBI-1) in the pre- and post-event period.

Table 3
Results of the PCA analysis showing variables and corresponding factor loadings (F_L). Factor loadings ≥0.5 marked in bold.

Variables	Period	Factors		
		F1	F2	F3
SSI-1	Pre-event	−0.5	**0.7**	
Q_{avg}		**−0.92**	0.14	
Q_{max}		**−0.95**	0.22	
Q_{min}		**−0.75**	−0.04	
RBI		**−0.76**	0.32	
CvQ		**−0.83**	0.34	
T_{max}		**−0.8**	**−0.58**	
T_{min}		**−0.82**	**−0.55**	
T_{avg}		**−0.82**	**−0.56**	
VQ		**−0.75**	0.28	
P		**−0.66**	−0.03	
Eigenvalue		6.82	1.8	
Accounted variance (%)		62	16.4	
Cumulative (%)		62	78.4	
SSI-1	Post-event	−0.39	**−0.63**	**−0.52**
Q_{avg}		**−0.92**	−0.09	−0.3
Q_{max}		**−0.95**	−0.22	−0.09
Q_{min}		**−0.73**	0.19	**−0.58**
RBI		**−0.81**	−0.35	0.28
CvQ		**−0.79**	−0.44	0.38
T_{max}		**−0.79**	**0.57**	0.14
T_{min}		**−0.83**	**0.53**	0.08
T_{avg}		**−0.81**	**0.55**	0.11
VQ		**−0.66**	−0.44	0.41
P		**−0.84**	−0.03	−0.09
Eigenvalue		6.81	1.9	1.14
Accounted variance (%)		61.9	17.3	10.3
Cumulative (%)		61.9	79.2	89.5

Note. SSI-1 – Standarised Streamflow Index, Q_{avg} – average discharge, Q_{max} – maximum discharge, Q_{min} – minimum discharge, RBI – Richard-Backer Index, CvQ – coefficient of variability for discharge, T_{max} – maximum air temperature, T_{min} – minimum air temperature, T_{avg} – average air temperature, VQ – quartile variation coefficient for discharge, P – precipitation.

temperatures inhibit the snowmelt and replenishment of soil moisture deficit, thus limiting streamflow variability. Time-averaged wavelet power spectra for the pre- and post-event period showed that the duration of high-frequency periodicity in streamflow significantly increased from 8 to 11 days in the pre-event period up to 3–11 days in the post-event period (Fig. 6). The increase in periodicity at shorter period bands in the post-event period was mostly confined to the hydrologically active season and reflected the short-term, intermittent nature of streamflow variability. This type of oscillation could imply a greater sensitivity of streamflow to prevailing meteorological conditions.

4.6. Changes in streamflow response to meteorological conditions

Precipitation and streamflow were strongly coherent at ~1 year period band during the entire study period. Warm colours denote a high degree of linear correlation between cycles, while the right-pointing arrows indicate an in-phase oscillation between precipitation and streamflow. The streamflow in the post-event period became more responsive to precipitation at shorter time scales (3–16 days) in the hydrologically active season (Fig. 7A), especially in midsummer when the catchment was wet following snowmelt and rainfall that generated a rapid flow response. This type of short scale periodicity has, however, an intermittent character, yet the signals were not directly in-phase, implying that streamflow lagged precipitation. The short-scale coherence between precipitation and streamflow was markedly weaker during winter as the soils dried and froze. The increasing coherence between precipitation and discharge on short time scales was especially visible in the first 4 years (2014–2017) after the catastrophic windfall. In the following years, the short-scale periodicity has gradually weakened possibly due to the succession of new vegetation. Another strong coherence between precipitation and streamflow occurred at ~64 days period band. The direction or arrows at this period band indicates that oscillations were in-phase in the pre-event period. Though, in the first 4 years after the catastrophic windthrow streamflow either lagged or lead precipitation at this period band. The phase of common oscillation stabilised in 2019, about 6 years after the windthrow. It is also visible that the coherence between precipitation and streamflow decreased at low frequencies (in the 2-year period band) in the post-event period, indicating a long-term change.

Streamflow was strongly coherent with air temperature on an annual

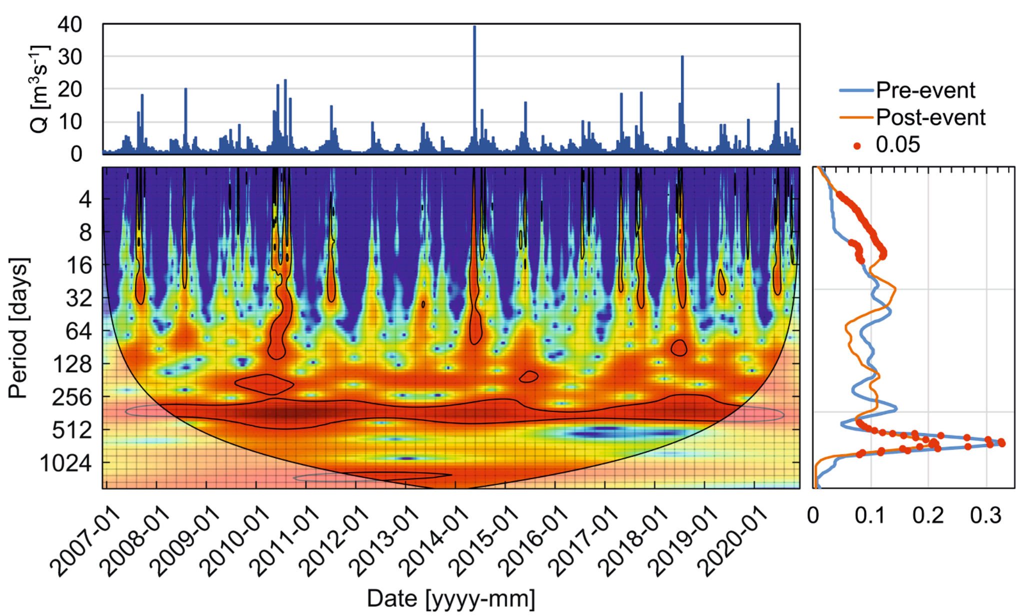

Fig. 6. Wavelet power spectrum of streamflow time series and time-averaged wavelet power spectra (right window) for the pre- and post-event periods with the corresponding 95% confidence levels (red points). Colours indicate the strength of the coherence, with black contour lines indicating significant coherence at the 5% significance level against the red noise. Lighter area indicates the cone of influence of the wavelet spectra which is influenced by edge effects. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

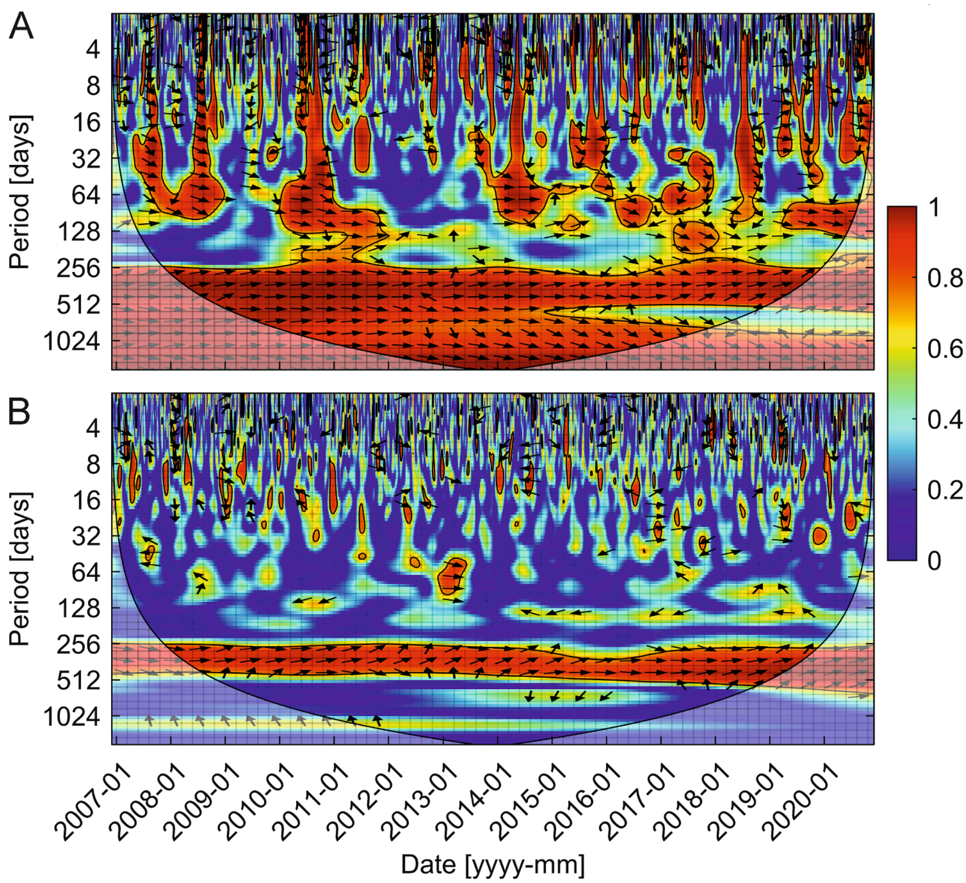

Fig. 7. Wavelet coherence between precipitation totals in Zakopane and streamflow (A), and average air temperature and streamflow (B). Colours indicate the strength of the coherence, with black contour lines indicating significant coherence at the 5% significance levels against the red noise. Lighter area indicates the cone of influence of the wavelet spectra which is influenced by edge effects. Arrows show the relative phase relationship between time series (in-phase, arrows point right; antiphase, arrows point left). Directions of the arrows indicate the degree to which the P and Q series are in-phase. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

time scale. Small patches of high coherence were also observed on short time scales (<1 month), though the coherence had an intermittent character and did not display a persistent pattern of co-oscillation. The arrow directions indicate that oscillations were almost in-phase in the 1-year period band, while in shorter period bands (<1 month) both in-phase and antiphase oscillations occurred equally often. Visually, no significant change in periodicity occurred in the post-event period (Fig. 7B).

It should be noted, however, that the lack of strong coherence between streamflow and meteorological data might be associated with the distance between the catchment and meteorological station. Although small changes in precipitation can be neglected, changes in air temperature between a sheltered mountain valley and the town at the foreground of mountains could be substantial.

4.7. Changes in root zone soil wetness and its relation to streamflow

Changes in soil wetness are related to the intensity and distribution of precipitation throughout the year. However, they also change with depth in the soil profile. The short-term correlation with precipitation usually decreases with increasing depth, reflecting the growing role of long-term water storage. Thus, the analysis of the root zone soil wetness in the extended zone between the surface up to 100 cm depth allows one to track both long- and short-term changes in vegetation-accessible water storage associated with land use changes.

The results showed that the soil wetness in the root zone exhibited several periodicities during the analysed time period. The most persistent was the annual periodicity that reflects the changes in soil wetness over the course of the year, with the highest soil wetness occurring during snowmelt (April). Low soil wetness usually occurs during summer, which in fact is the period with the highest amounts of rainfall in the Tatras. The exceptionally low soil wetness occurred more frequently in September after a prolonged period of dry weather. The low wavelet power in the wavelet power spectrum coincided with a country-scale drought in 2012 that affected about 71% of the country (Pociask-Karteczka et al., 2018). Fortunately, the drought occurred in months beyond the maximum growth of the trees. In turn, high wavelet power in the wavelet power spectrum coincided with flood events, i.e., in 2010, when large parts of the country were flooded or in 2014 when the flood hit the Carpathian region. Other significantly correlated, but intermittent, cycles were observed at about 3 months. This cycle is associated with the rainfall period during summer months and was more evident in the pre-event period. It should be noted that the 3-month periodicity completely vanished during the drought event in 2012 and was remarkably weaker after deforestation (Fig. 8A). The 3-month periodicity became stronger about 6 years after the catastrophic windfall, which might indicate a possible hydrological recovery time. The lack of a strong periodical pattern during this time indicates that soil wetness in the root zone was quite homogeneous throughout the year. During this time, a short-scale periodicity of about 2 weeks in root zone soil wetness became more apparent, especially between July and September in the post-event period, approximately from 2016 until the end of analysed time series. This periodicity has, however, an intermittent character and low power indicating smaller amplitudes of

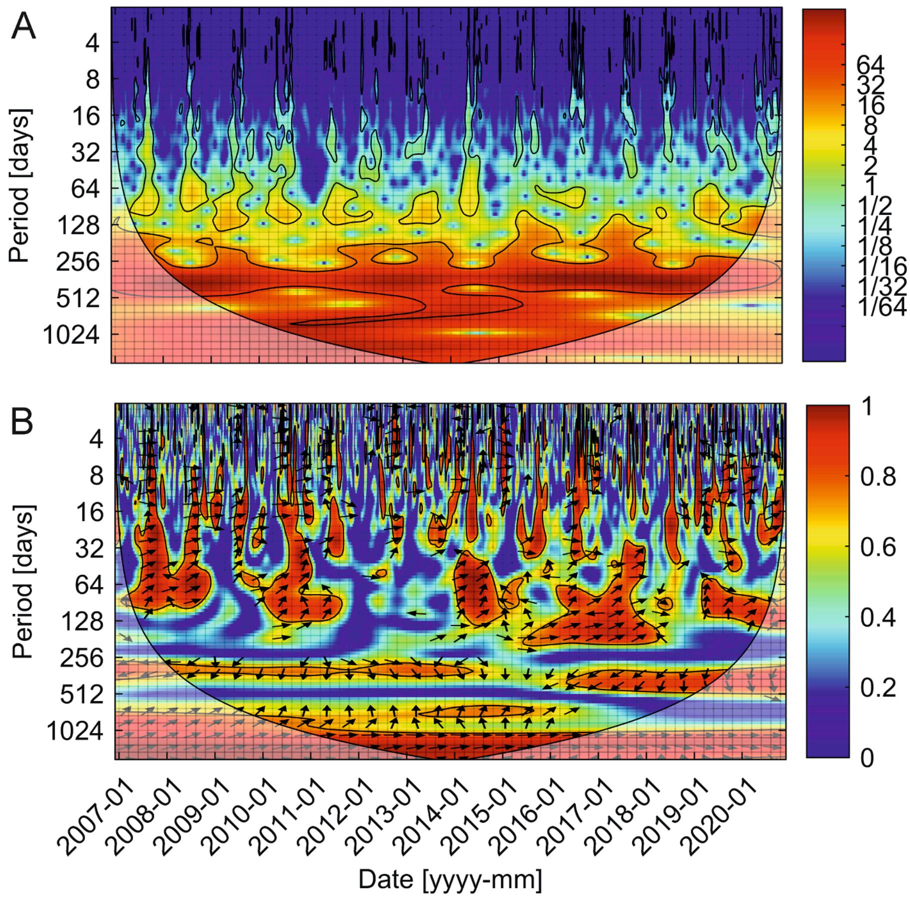

Fig. 8. Wavelet Power Spectrum of the root zone soil wetness (A) and Wavelet Coherence between the soil wetness and streamflow. Colours indicate the strength of the coherence, with black contour lines indicating significant coherence at the 5% significance levels against the red noise. Lighter area indicates the cone of influence of the wavelet spectra which is influenced by edge effects. The arrows show the relative phase relationship between time series (in-phase, the arrows point right; antiphase, the arrows point left). The directions of arrows indicate the degree to which the soil wetness and streamflow time series are in-phase. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

oscillations.

Root zone soil wetness was strongly coherent with streamflow in the extended period band from about 1 day to 3 months, on an annual time scale and at lower frequencies (≥3 years). The highest coherence was observed at the annual time scale. Notably this coherence clearly attenuated after deforestation (2014–2016) for about 2 years, resulting in a 2-year cycle between 2014 and 2015. Significant coherence was observed at about 3-month period band. The right-up pointing arrows indicate that the discharge was occasionally leading the root zone soil wetness. At short scale (<1 month), the coherence of intermittent character was more evident in the post-event period. Strong coherence was observed on longer time scales (~half year) after the catastrophic windfall. Right-pointing arrows at short (<1 month) and long (>3 years) period bands indicate that discharge was in-phase with the root zone soil wetness.

5. Discussion

The inherent noise in hydrological time series accompanied by growing non-stationarity in long time series requires alternative methods to understand the functioning of the catchment and to assess the potential susceptibility of the catchment to catastrophic events, land use changes and short-term climatic patterns. Here, we applied the concept of streamflow flashiness and hydrological drought to investigate hydrological changes caused by the catastrophic windfall. Some of the results obtained from this study agree with the current knowledge about forest-hydrology disturbances in temperate climate zone; however, a part of the results brings some new aspects that have not been considered so far and could be addressed in the future studies.

5.1. Soil wetness in the root zone

Temporal changes in the soil wetness are strongly correlated with the distribution of precipitation over the year, therefore, are more evident in climatic zones with pronounced rainfall seasonality (Gao et al., 2014). However, the correlation decreases with depth, which is especially pronounced in forested areas (Liu et al., 2017). Our results revealed that the catastrophic windfall hindered the intra-annual periodical patterns of soil wetness in the root zone, causing an increase in high-frequency (~2 weeks) periodicity and a decrease in the mid-frequency (~3 months) periodicity. The decrease in mid-frequency periodicity after deforestation might indicate less water storage in the root zone. Typically, shallow soil layers are dominated by high-frequency periodicity (<1 year) and are deprived of low-frequency periodicity owning to limited water storage and stronger correlation with precipitation. Conversely, deeper soil layers are dominated by mid- and low frequency periodicity reflecting long-term water storage and are less responsive to rainfall events. The soil wetness extending up to 100 cm, representing the maximum thickness of soil in this area, incorporates both long- and short-time oscillations.

Satellite-derived multi-layer soil moisture data are commonly employed to assess drought severity, due to their strong correlation with meteorological drought indices, such as Palmer Drought Severity Index and SPI (Sehgal and Sridhar, 2019; Sehgal et al., 2017). Research indicates that total column soil moisture, observed over extended aggregation periods, exhibits a strong correlation with long-term drought indices, while surface soil moisture, analysed over shorter aggregation periods, is correlated with short-term drought indices. It should be noted, however, that the response of soil profile to drought is influenced by various factors, including the composition and texture of soil, topography, and vegetation (Sehgal and Sridhar, 2019).

The shallow Eutric Cambisols and Cambic Rendzic Leptosols prevalent in the catchment are typically well-drained soils, allowing for the efficient drainage of excess water. This characteristic can influence the root zone soil moisture, ensuring that it sustains an optimal moisture level for plant growth without excessive waterlogging. Despite being well-drained, these soils can still retain a significant amount of water in the root zone. This water retention capacity is influenced by the soil's texture, structure, and organic matter content and thus, it may undergo changes associated with land use changes and vegetation cover. For instance, windthrow soils often exhibit significantly higher field moisture in O horizons compared to forest soils (Wasak et al., 2020).

The research suggests that vegetation plays a key role in the partitioning of water between runoff and transpiration in the root zone (Donohue et al., 2012; Nijzink et al., 2016; Zhang et al., 2023). Previous works reported a strong correlation between the root zone storage capacity and runoff coefficients (Gao et al., 2014). Moreover, significant changes in runoff coefficients were often observed after the removal of vegetation. The research also showed that a decrease in the root zone storage capacities in deforested catchments is followed by its gradual recovery towards a new equilibrium (Nijzink et al., 2016). The research suggests that increasing streamflow after deforestation is accompanied by a corresponding reduction of evaporative fluxes, which in turn could be linked to the diminishing catchment-scale water storage in the unsaturated soil that is accessible for the transpiration of vegetation (Hrachowitz et al., 2021).

Arguably, the weakening of 3-month periodicity in the root zone soil wetness in the post-event period might indicate a decrease in groundwater fluctuations. Previous studies suggest that roots follow groundwater table fluctuations by increasing rooting depths when water table drops (Tron et al., 2015). The rooting depth is considered as a major control of groundwater recharge. Previous research showed that the deep-rooted vegetation may significantly reduce groundwater recharge (Li et al., 2019). Our study revealed that the 3-month periodicity associated with long-term water storage became stronger about 6 years after the windthrow. This may suggest a potential recovery time linked to the succession of new vegetation. Studies conducted in other mountain regions in Europe have shown that excavated pits can become fully covered with vegetation within 5 to 10 years following a forest disturbance (Fischer et al., 2002; Wohlgemuth et al., 2002). In the initial stages of succession, deforested and unburnt areas are colonised by plants from the soil seed bank or by those surviving through the root system, such as grasses and herbs, followed by shrubs and trees (Šoltés et al., 2010). The recovery after deforestation, however, can vary widely depending on various factors, including the extent of the deforestation, the type of ecosystem, climate, and human interventions.

Soil structure damaged by tree-throw and detached root plates on deforested hillslopes may increase surface wash and throughflow. Surface runoff in forested catchments is relatively rare and is usually limited to steep slopes or extreme rainfalls. As the result, the formation of runoff in forested areas is dominated by infiltration, water storage and preferential flow (Reinhardt-Imjela et al., 2018). Root channels are the primary paths for preferential flow of storm water and solute transport in forested hillslopes (Luo et al., 2019; Sidle et al., 2001). Preferential flow refers to fast water flow through a small fraction of pore spaces and typically occurs during infiltration (Beven and Germann, 2013). The frequency of preferential flow varies temporally and spatially. Preferential flow is less frequent during spring and more frequent during summer and early autumn, probably due to water repellency (Demand et al., 2019). Previous research showed that deforestation may increase the frequency of preferential flow for events with identical total precipitation owning to higher soil moisture (Liu and Lin, 2015). The latter provides the connection between preferential flow paths (Sidle et al., 2001) and increases the subsurface flow response of the hillslope (Sidle et al., 2000). The increase in preferential flow may in turn, result in the enhanced export of nutrients and Dissolved Organic Carbon to groundwaters after deforestation (Wiekenkamp et al., 2020). Previous studies in the Kościeliska Valley showed an increasing concentration of nitrates in deforested sub-catchments due to lower nutrient uptake by plants (Żelazny et al., 2018), which might serve as evidence of increasing frequency of preferential flows. Higher antecedent moisture conditions and the lack of interception in deforested catchments may as well result in a higher frequency of slower sequential (piston) flow (Wiekenkamp et al., 2020).

Windthrow pits can intercept subsurface flow leading to the intensification of gully erosion, landslides and the transport of fine sediment, bed load and woody debris into the river channel. Although the geological and soil structure of the Tatra Mountains does not make landslides as prevalent as in the Outer Carpathians, previous research in the catchment documented multiple shallow landslides following the catastrophic windthrow (Strzyżowski et al., 2021). The occurrence of such processes is usually favoured by high precipitation totals, mountainous relief, and land use changes, as well as the Topographic Wetness Index and soil type, all of which contribute to slope instability (Sujatha and Sridhar, 2021).

5.2. *Windthrow impact on streamflow flashiness*

Our results showed that the catchment is much more responsive to precipitation in the post-event period with less catchment storage that would mediate flow variability and flashiness of the streamflow. The flashiness of streamflow increased significantly in September in the post-event period, which might be related to a faster creation of surface runoff from neighbouring slopes due to limited infiltration and/or water storage capacity.

Deforestation strongly affects soil surface features, which in turn determine whether the rain infiltrates or create the runoff. Previous research showed that stream waters in the post-deforestation period tend to have slightly higher mean fractions of young water than in the pre-deforestation period (Hrachowitz et al., 2021), which is ascribed to an increasing portion of the surface runoff. In temperate climate zone, surface runoff is dominated by Hewlettian/Dunnian flow (saturation excess overland flow), which is associated with limited storage of surface soils, and it typically creates subsurface flow that sustains the baseflow in streams and groundwater recharge (Huxman et al., 2005). In turn, the Hortonian overland flow (infiltration excess runoff), which is favored by diminishing infiltration capacities resulting from compaction of the soil may lead to reduced groundwater recharge and baseflow (Peña-Arancibia et al., 2019). The studies suggest that land degradation (Descroix et al., 2007) and harvesting operations (Croke et al., 1999) may promote the occurrence of Hortonian overland flow. These types of runoffs usually act at different time scales, i.e., the Dunnian flow is slower than Hortonian flow but faster than the baseflow. If we hypothetically assume that these types of flows might be apparent in the streamflow hydrograph, the shifts in high frequency periodicity of streamflow might be useful to identify these mechanisms. Our results, indeed, showed an increase in high frequency periodicity in streamflow, which might be associated with the mechanism of surface runoff generation, i.e., higher frequency of Hortonian and Dunnian flows.

The formation of overland flow may be favored by prolonged periods of drought events followed by intensive rainfalls. Soils in the catchment are relatively shallow; therefore, even several days of warm weather may result in the rapid desiccation of unvegetated soils. Previous research conducted in the neighboring catchment showed an increase in soil temperature in the summer months in deforested catchments. Furthermore, soil temperature on deforested slopes was strongly correlated with air temperature, while diurnal fluctuations in soil temperature on deforested slopes were up to 3 times higher than on forested slopes (Siwek, 2020, 2021), indicating a higher vulnerability to desiccation.

Typically, unvegetated and desiccated soils tend to become hydrophobic after prolonged period of meteorological droughts, owning to soil–water hysteresis (Haines, 1930). Hysteresis occurs because the volumetric water content on a drying curve is always higher than that on a wetting curve for the same matric suction (Sławiński, 2011). This effect might lead to dangerous flash floods even during droughts, when heavy rain events hit super-dry soils. In fact, the Carpathians, including the Tatra Mountains, are among the places particularly affected by flash floods (Ballesteros-Cánovas et al., 2015; Bryndal et al., 2017; Pociask-Karteczka et al., 2018).

5.3. Windthrow impact on streamflow characteristics

Our results showed that the catastrophic windfall caused the increase in low flows in all months and less frequent hydrological droughts. This increase was particularly higher at the end of the growing season. Earlier research concerning the impact of deforestation on streamflow most frequently reported either the increase of low flows (Fleischer et al., 2020; Zhang and Wei, 2012) or the lack of substantial change in streamflow (Coble et al., 2020). The impact of deforestation on hydrological conditions manifests not only in the amount of flowing water but also in seasonal changes in the runoff. The research in the Slovak Tatra Mts. reported an increase in average and low monthly flows after deforestation in all months of the year as well as the shift of seasonal high flows to the summer period (July-August) (Fleischer et al., 2020). Another study in Sweden reported about 35% increase in catchment runoff after deforestation (Sørensen et al., 2009). According to this research, the increase of runoff was greater (58–99%) during low flows in the growing season than during average flows in the non-growing season (11–34%).

Deforestation may affect dry season flows by decreasing interception and evapotranspiration losses, which in turn decrease soil moisture deficit in the dry period. This decrease might lead to elevated dry season flows where at least part of the dry season flow is derived from the soil moisture reservoir. The exceptionally large increase in low flows in deforested catchments is ascribed to the reduction of transpiration. Elevated low flows often persist for a short time after deforestation, though after the stabilisation of vegetation, a drop in low flows to levels even lower than before are frequently observed. The decrease in low flows after the regrowth of vegetation is attributed to plant species that often exhibit a higher level of transpiration (Moore et al., 2004).

Earlier studies showed that the removal of tall vegetation results in less water demand due to reduced evaporative losses, which in turn accelerates water circulation in the catchment leading to increased volumes of streamflow (Bosch and Hewlett, 1982; Brown et al., 2005). The increase in flow is at the same time favoured by the reduced catchment storage. This is because the intact forest act as a sponge absorbing and storing water and releasing it slowly during the subsequent dry season (Peña-Arancibia et al., 2019). The water storage potential of forest depends, however, on the type of vegetation. Deep-rooted trees are characterised by higher water demand than shallow-rooted plants (Jackson et al., 2005). Water demand by trees decreases also with the age. Young trees typically need more water for growth. Species that naturally inhabited deforested areas are usually more resistant to prevailing weather conditions owning to well-developed horizontal and vertical structure, which in turn, might favour the storage of water in the root zone (Schüler, 2006). More recent studies, however, have shown that summer rainfall may have little impact on soil water storage that provides water for tree growth and may not be the main source of water in midsummer transpiration (Allen et al., 2019). Deciduous trees like beech and oak tend to use winter precipitation, whereas spruce uses water of more diverse seasonal origins, which might potentially mitigate its vulnerability to summer droughts (Allen et al., 2019).

In forested catchments only a part of rainwater reaches the ground because a large part of it is retained in tree crowns, evaporates back to the atmosphere or is transpired causing interception losses. These losses affect significantly not only the water budget but also the climate by removing about a quarter of annual precipitation from the terrestrial hydrologic cycle (Dingman, 2002; Návar, 2020). The interception significantly reduces the amount of water flowing as surface runoff. Interception potential increases with the area of forest. However, the intensity of interception depends on different factors, including the timing and intensity of rainfall, vegetation structure, and meteorological conditions (Dingman, 2002). Previous research reported that interception of precipitation by spruce increases with the stand age and its dimension, ranging between 16 and 48% (Šrámek et al., 2019).

Forests in particular have a positive effect on rainwater infiltration by increasing infiltration capacity and moisture capacity, thereby reducing surface runoff (Pea-Arancibia et al., 2019). It is commonly believed that deforestation has the greatest impact on dry season flows, which is often equated with the summer season. In streams of the Tatra Mountains, low flows occur most frequently in winter when significant water masses are stored in snow cover, and precipitation totals are lower than in the summer months (Żelazny et al., 2019b). Our results showed that the severity of droughts decreased after the catastrophic windfall, and droughts became less frequent, which is directly associated with increased low flows, as indicated by PCA analysis. Observed changes in low flows may in turn be associated with the baseflow and the soil–water storage capacity of the catchment.

5.4. Windthrow impact on snow cover and snowmelt

Significant deforestation can affect the snow retention and the duration of snow cover, thereby influencing the intensity of spring snowmelt. Research on deforestation in temperate cool climate catchments often reports an increase in average annual flows and the increase in peak and high flows during spring snowmelt. This increase is plausibly associated with more intensive snowmelt due to higher exposure of cold, open snowpacks to winter sunlight, resulting in the rapid formation of surface runoff. Some earlier studies reported that although forest clearing accelerated the onset of the main thaw period, the snowmelt rate was insufficient to produce significantly higher flow in the early snowmelt season (Kuras et al., 2012). Other studies, however, reported not only an earlier snowmelt but also about a 60% increase in spring flows (Macdonald et al., 2003). Our results, in turn, revealed a significant increase in streamflow flashiness, as well as the increase in monthly maximum and average flows during winter thaws, and an increase in low flows during the snowmelt. The exposed snowpack is plausibly more sensitive to daily amplitudes of air temperature. Winter daytime temperatures were often above 0°C, resulting in significant snowmelt, while nighttime temperatures led to the refreezing of snowpack. The snowpack dynamics likely contributed to the observed increase in average and high flows, increased streamflow flashiness as well as the reduction in drought events. These findings align with previous studies indicating that snowmelt on frozen soils during mid-winter thaw events (in February) and late-season snowmelt (April–May) in the interim of the northern hemisphere continents can cause or exacerbate major floods (Hillard et al., 2003).

Previous research suggests that snow ablation in open areas could be almost two times faster than in forested sites and about 18% faster after forest defoliation due to a bark beetle outbreak (Jenicek et al., 2018). The research showed that the duration of snow cover is about 11 days shorter on mountain tops with decayed forests than in forested areas (Hesslerová et al., 2018). In forested catchments, snow cover usually lasts much longer than in the open areas, while the snowmelt rates can be up to 70% lower compared to open areas, which is related to the attenuation of incoming shortwave radiation by the tree crowns

(Varhola et al., 2010). The melting rate of snow cover in the forest is also modified by reduced sensible and latent heat fluxes due to the reduced wind speed by forest canopy (Varhola et al., 2010). Observations suggest that up to 45% of the annual snowfall can sublimate from coniferous canopies in cold, dry continental environments (Essery and Pomeroy, 2001). In forested catchments, snowmelt beneath the canopy is dominated by long-wave radiation, while in harvested areas, the snowmelt is dominated by short-wave radiation and turbulent heat fluxes, resulting in a higher magnitude and frequency of snowmelt floods (Essery et al., 2003; Green and Alila, 2012).

Our results showed that low flows significantly increased in April, which overlaps with the snowmelt period, but this increase has a gradual character. Our results are in alignment with earlier research reporting an increase in peak flows in deforested catchments resulting from higher snow accumulation and limited interception of snow by the forest canopy (Moore and Wondzell, 2005; Pomeroy et al., 2012). Studies suggest that depending on the canopy structure, the Soil Water Equivalent (SWE) in open areas could be by up to 40% higher than in forested sites (Jenicek et al., 2018). Higher accumulation of snow during winter in open areas often results in larger volumes of snowmelt (Gelfan et al., 2004). Studies from the Slovak Tatra Mountains suggests, however, that higher snow depth and SWE in disturbed forest have not significantly affected the volume of snowmelt (Bartík et al., 2019). Moreover, the studies suggest that snow accumulation in open, wind-exposed areas is less than in coniferous forests (Pomeroy and Gray, 1995). According to this study, in open wind-exposed areas up to 75% of the annual snowfall can be transported and sublimated by blowing snow (Pomeroy and Gray, 1995). Furthermore, owning to the mid-winter thaws that are especially pronounced in open areas, the SWE before spring snowmelt can be greater in forested catchments (Murray and Buttle, 2003).

The increase in low flows during winter may prevent ice formation and create a more stable environment for aquatic life. On the other hand, increasing streamflow in winter may pose a risk of water shortages in the later part of the year, especially during the growing season when vegetation is more active and evaporative demand is maximal (i.e., Rood et al., 2008). Considering deforestation, limited water resources in the later part of the year, during the growing season, may favour the succession of xerothermic plant species that are more resistant to droughts. Our analysis revealed, however, that low flow has also increased in the growing season, which was partially compensated by the decreasing long-term water storage in the root zone and potentially lower evapotranspiration rates, since herbs and grasses that colonise the slopes in the first stage possess a much lower capacity for evapotranspiration than trees.

6. Conclusions

In this study, we analysed the impact of catastrophic windfall on streamflow characteristics using a paired-year approach. Periodicity in streamflow time series was tracked in relation to meteorological conditions (precipitation, air temperature) and to subsurface soil water storage using remotely sensed soil wetness in the root zone. The results revealed that the flow regime and soil wetness in the root zone were significantly altered by deforestation, and hydrological recovery from these alterations through the succession of new vegetation took several years. Increasing streamflow, streamflow flashiness, and periodicity indicate possible changes in the mechanism of surface runoff generation. Changes in soil wetness in the root zone associated with vegetation growth reflect changes in subsurface storage-streamflow dynamics and the mechanism of subsurface water flow. The magnitude of changes decreases with time, possibly due to succession of new vegetation and the consolidation of slopes by plants. However, the catastrophic windfall tends to reconfigure the periodical patterns of variability and scales of common oscillations. Catastrophic windthrow affected mostly low flows and, to a lesser extent, average and high flows. This increase was most significant in the growing season, which could be associated with vegetation removal, and thus, decreased infiltration, evaporation, interception, and accompanying decrease in soil water storage. Hydrological droughts became less severe and less frequent during the year. The substantial increase in average and maximum flow, as well as in flashiness during winter, indicates that deforestation significantly affects the accumulation and rate of snowmelt in sun-exposed areas, and to a lesser extent, the volume of snowmelt. On average, the changes that occurred in low flows and drought frequency accounted for about 10% of the observed variability in streamflow and are largely driven by deforestation, suggesting that the storage-streamflow relationship is one of the most important mechanisms for streamflow generation under ongoing land use changes. Hydrological changes that occur after natural forest disturbances that manifested in rapid water circulation in the catchment, higher baseflow, and limited long-term soil–water storage, may contribute to the regrowth of diversified plant species, which are more resistant to the prevailing conditions. The result obtained from these studies may advance our understanding and improve our ability to predict the effects of vegetation changes on water resources.

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CRediT authorship contribution statement

Agnieszka Rajwa-Kuligiewicz: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Anna Bojarczuk:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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